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Results from the PRIMA Trial: Comparison of the STHLM3 Test and Prostate-specific Antigen in General Practice for Detection of Prostate Cancer in a Biopsy-naïve Population

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Abstract

Background: Current management of prostate cancer (PC) lacks biomarker tests and diagnostic procedures that can accurately distinguish clinically significant and clinically insignificant PCs at an early stage of the disease.

Objective: To compare the Stockholm 3 (STHLM3) test and prostate-specific antigen (PSA) as entry tests for magnetic resonance imaging (MRI) in a prospective study of PC diagnosis in general practice.

Design, setting, and participants: Participants were biopsy-naïve men aged 50–69 yr who had a PSA test in general practice. Participants with PSA 1–10 ng/ml also had an STHLM3 test and were referred for MRI if the STHLM311 test was positive (risk $\geq 11\%$) and/or PSA ≥ 3 ng/ml, and to targeted MRI-guided biopsy (MRGB) if their Prostate Imaging-Reporting and Data System (PI-RADS) score was ≥ 3 .

Outcome measurements and statistical analysis: The primary outcome was the number of International Society of Urological Pathology grade group ≥ 2 (GG ≥ 2) cases detected with a positive STHLM311 test versus PSA ≥ 3 ng/ml. Post hoc analysis was performed using a higher STHLM3 test cutoff (risk $\geq 15\%$; positive STHLM3₁₅ test).

Results and limitations: Between January 2018 and December 2021, we recruited 1905 men. The STHLM3 test was performed in 1134 participants. Of these, 437 underwent MRI and 117 underwent MRGB, which detected 38 (32.5%) GG ≥ 2 and 52 (44.4%) with GG 1 cases. In comparison to PSA ≥ 3 ng/ml, a positive STHLM311 test increased detection of GG ≥ 2 from 30 to 37 cases (23.3%, 95% confidence interval [CI] 5.6–52.2%) and detection of GG 1 from 37 to 50 cases (35.1%, 95%CI 11.6–66.7%). STHLM3₁₅ positivity did not differ from PSA ≥ 3 ng/ml regarding detection of GG ≥ 2 PC (30 vs 32; 6.6%, 95% CI –8.1% to 25.9%), GG 1 PC (37 vs 37; 0.0%, 95% CI –19.6% to 25.0%), or MRGB use (88 vs 83;

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–5.7%, 95% CI –17.9% to 7.4%), but reduced MRI scans from 320 to 236 (–26.2%, 95% CI –33.1% to –18.9%).

Conclusions: The STHLM311 test improved sensitivity but not specificity for detection of GG ≥ 2 PC in the clinical setting of nonsystematic PC testing in general practice. Further studies are needed to validate a possible benefit of using a higher cutoff for STHLM3 positivity as an entry test for MRI.

Patient summary: We used a test called STHLM3 for detection of prostate cancer in general practice and compared its performance to the conventional PSA (prostate-specific antigen) test. We found that STHLM3 test results of 11% or above were not better at selecting men for MRI (magnetic resonance imaging) scans than the PSA test with a cutoff of 3 ng/ml or above. Analysis suggested that a higher cutoff for a positive STHLM3 test may improve selection of men for MRI scans, but further validation is needed.

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1. Introduction

Current management of prostate cancer (PC) lacks biomarker tests and diagnostic procedures that can accurately distinguish clinically significant and clinically insignificant PCs at an early stage of the disease, as only the former are likely to require diagnosis and treatment [1,2]. The number of men diagnosed with PC is expected to increase in the future, emphasizing the need for better diagnostic strategies [3].

The first steps for diagnosis of PC include a prostate-specific antigen (PSA) test and digital rectal examination (DRE), typically performed in primary care. However, PSA has low specificity and DRE has low sensitivity, leading to overdiagnosis of clinically insignificant PCs and underdiagnosis of clinically significant PCs [2,4,5]. Recent studies have shown that PC diagnosis can be improved by performing magnetic resonance imaging (MRI) and targeted MRI-guided biopsy (MRGB) [6–11]. Accordingly, MRI of the prostate is now recommended as a follow-up examination in cases with elevated PSA and/or abnormal DRE to determine the need for biopsy and to facilitate targeted biopsy of lesions suspicious for cancer [12–14]. This has put many radiology and urology departments under pressure owing to lack of MRI capacity and/or trained personnel. A better entry test than PSA before MRI could help to overcome this bottleneck.

Emerging blood-based tests, such as 4Kscore, the Prostate Health Index, and the Stockholm 3 (STHLM3) test [15], can improve selection of men for systematic prostate biopsy [16], but studies including MRI and targeted biopsy are scarce [17]. One exception is the STHLM3-MRI screening-by-invitation study, which compared STHLM3 and PSA as entry tests for biparametric MRI followed by MRGB and systematic biopsy [18]. The study reported that STHLM3 test scores $\geq 15\%$ could identify the same number of clinically significant PCs (defined as International Society of Urological Pathology [ISUP] grade group [GG] ≥ 2) as PSA ≥ 3 ng/ml, while avoiding 36% of MRI and 8% of biopsy procedures [18]. However, the results were based on recalibration of the cutoff for STHLM3 test positivity from the original of $\geq 11\%$ to $\geq 15\%$, and lack independent validation. Furthermore, systematic PC screening by invitation is controversial and not generally implemented [12–14,19] and

data are needed on the performance of STHLM3 as an entry test for MRI among men consulting their general practitioner (GP) because of health concerns or urologic symptoms indicative of PC.

Here we report results from the first prospective clinical diagnostic trial (PRIMA) comparing STHLM3 and PSA as the entry test for MRI in general practice as the entry point. The study thereby mimics the clinical workflow in many countries that have implemented an MRI-first strategy but do not recommend population-based screening. Our hypothesis was that in comparison to PSA measurement, the STHLM3 test would increase detection of GG ≥ 2 PC and reduce detection of GG 1 PC when using MRI and MRGB as confirmatory tests.

2. Patients and methods

2.1. Ethics considerations

This study conformed to the principles of the Helsinki Declaration [20] and was approved by The Central Denmark Region Committee on Health Research Ethics (reference number 1-10-72-245-17). The Danish Data Protection Agency (reference number 1-16-02-719-17) was informed about data collection. All participants received written and oral information about the study and gave written consent. The study was registered at ClinicalTrials.gov (NCT03431753). This report follows the Standards for Reporting of Diagnostic Accuracy Studies (STARD) 2015 guideline [21].

2.2. Setting and participants

We conducted a prospective diagnostic study (termed PRIMA) with a paired, test-positive design in a collaborative effort between primary care and hospitals in the Central Denmark Region (1.3 million inhabitants and 352 general practices). All Danish citizens have free access to GPs, who serve as gatekeepers to the rest of the health care system [22]. A total of 148 (42.0%) general practices participated in this study, together with one radiology department, one pathology department, and two urology departments.

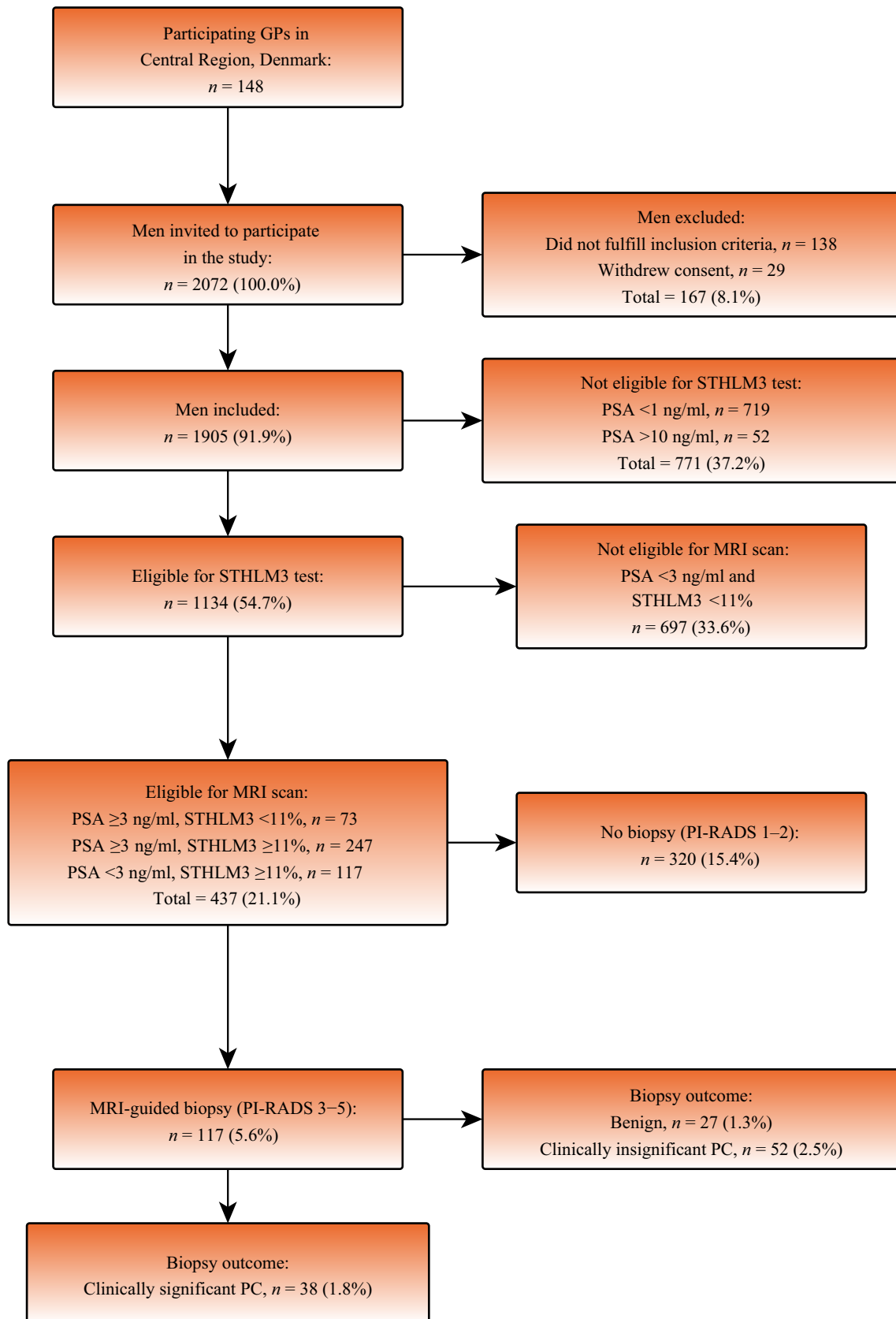


Fig. 1 – Flow diagram of participants included in the study. PSA = prostate-specific antigen; GP = general practice; MRI = magnetic resonance imaging; PC = prostate cancer; PI-RADS = Prostate Imaging-Reporting and Data System.

2.3. Inclusion

The patient flow is presented in [Figure 1](#). Patients were invited to participate if, in agreement with their GP, they had a PSA test and met the following eligibility criteria: age 50–69 yr, no prior diagnosis of PC or other urogenital cancer, no prior prostate biopsy, no prior elevated PSA, negative DRE, able to understand and speak Danish, and no contraindication to MRI. The patients formed a consecutive series and were recruited between January 1, 2018 and December 31, 2021. We restricted the study to men aged 50–69 yr as this was the age group investigated in the original STHLM3 study [15].

2.4. Blood tests

Eligible men had a PSA test and were asked about their family history of PC (father, brothers, sons) and their use of 5 α -reductase inhibitors (yes/no/unknown within 3 mo). The participants also donated blood samples for the STHLM3 test, which was performed for participants with PSA between 1 and 10 ng/ml (diagnostic grey zone for PSA). In accordance with the test definition, the STHLM3 test is reported as negative if PSA is <1.0 ng/ml [15,18]. Hence, men with PSA <1 ng/ml were excluded from further analyses in the current study. Men with PSA >10 ng/ml were also excluded from further analyses in the study and received follow-up at one of two participating urology departments at Aarhus University Hospital (AUH) or West Jutland Regional Hospital, Holstebro (RHH) according to clinical guidelines in Denmark.

The STHLM3 test was performed at A3P Biomedical (Uppsala, Sweden) and results were reported as a risk score percentage, reflecting the estimated risk of GG $\geq$ 2 PC. An STHLM3 risk score $\geq$ 11% was reported as a positive STHLM311 test [18].

2.5. Patient workup

There was no further workup for participants with a negative STHLM311 test (risk score <11%) and PSA <3 ng/ml. Participants with a positive STHLM3₁₁ test and/or PSA $\geq$ 3 ng/ml were referred for multiparametric MRI (mpMRI) at the Department of Radiology, AUH. The mpMRI was performed using a 3-T platform (Skyra, Siemens Erlangen, Germany) and a radiologist certified in MRI of the prostate scored each lesion in accordance with the Prostate Imaging-Recording and Data system (PI-RADS) classification [23,24]. The radiologist was aware of the result of the PSA test.

Participants with a PI-RADS score of 3–5 underwent in-bore MRGB at the Department of Radiology, AUH [25]. Biopsy tissue was handled in accordance with clinical routine at the Department of Pathology, AUH, and classified according to ISUP grade group [26]. We defined clinically significant PC as ISUP GG $\geq$ 2 and clinically insignificant PC as ISUP GG 1. Participants with a PI-RADS score of 1–2 and PSA $\geq$ 3 ng/ml, regardless of the STHLM311 test result, received follow-up at one of the two participating urology departments. Participants with a PI-RADS score of 1–2, PSA <3 ng/ml, and a positive STHLM3₁₁ test received no further workup, but annual PSA testing was recommended.

After MRGB, participants were referred to a urology department (AUH or RHH) for treatment in the case of PC or for further workup in the case of a negative biopsy. The clinical outcome for each patient was recorded from patient files and grouped as (1) radical treatment, (2) active surveillance (AS), or (3) discharged to GP care.

2.6. Outcome measures

The primary outcome was the ability of PSA $\geq$ 3 ng/ml versus STHLM311 positivity to identify men with clinically significant PC (GG $\geq$ 2). Secondary outcomes were the number of MRI scans, biopsies, clinically insignificant PC (GG 1) cases detected, and benign biopsies, as well as clinical outcomes.

We also performed a post hoc analysis for all outcome measures using an adjusted cutoff for STHLM3 test positivity (risk score of $\geq$ 15%, referred to as a positive STHLM315 test) as this cutoff had the same sensitivity as PSA $\geq$ 3 ng/ml for detection of GG $\geq$ 2 PC in the MRI-STHLM3 screening study [18]. Finally, we performed sensitivity analyses for STHLM3 risk scores in the interval from 11% to 20%.

2.7. Statistical analysis

An a priori power calculation showed that 1500 participants should be recruited to obtain at least 90% power (α = 0.01) to show that the STHLM311 test was 50% more specific at the same level of sensitivity as PSA $\geq$ 3 ng/ml for diagnosis of GG $\geq$ 2 PC [27].

Data were analyzed using R v4.1.1 with RStudio v1.4.1717 (R Foundation for Statistical Computing, Vienna, Austria). The frequency and proportion are reported for categorical variables, and the median and interquartile range (IQR) for continuous variables. A χ^2 test was used to assess differences in distributions. To compute confidence intervals (CIs) when comparing positive STHLM3 and positive PSA tests, we used 1000 bootstrapped data sets using the R package *boot* v1.3-28.1. Receiver operating characteristic (ROC) curve analysis was performed using the R package *pROC* v1.18.0, and differences in area under the ROC curve (AUC) were tested via a paired Delong test. Statistical significance was set at p < 0.05.

3. Results

3.1. Trial population

Between January 1, 2018, and December 31, 2021, we recruited 2072 men in 148 general practices in the Central Denmark Region. Data collection closed on August 25, 2022. Of the 2072 participants, 138 (6.7%) were excluded because they did not meet the inclusion criteria and another 29 (1.4%) withdrew consent. Characteristics of the remaining 1905 participants are summarized in [Table 1](#).

The STHLM3 test was performed for 1134 participants with PSA between 1 and 10 ng/ml. Of these, 364 (32.1%) had an STHLM3 risk score $\geq$ 11%, corresponding to a positive STHLM311 test, and 770 (67.9%) had an STHLM3 risk score <11% ([Table 1](#)).

Consequently, 437 participants underwent MRI because of PSA $\geq$ 3 ng/ml and/or STHLM311 test positivity ([Fig. 1](#)).

Table 1 – Cohort characteristics

Parameter	Overall cohort (n = 1905)	PSA ≥ 3 ng/ml (n = 320)	STHLM3 positive	
			Risk $\geq 11\%$ (n = 364)	Risk $\geq 15\%$ (n = 236)
Median age, yr (IQR)	61 (56–66)	64 (60–67)	65 (61–68)	65 (61–68)
Age group, n (%)				
50–59 yr	811 (43)	77 (24)	83 (23)	49 (21)
60–69 yr	1094 (57)	243 (76)	281 (77)	187 (79)
FDR with PC (self-reported), n (%)				
Yes	455 (24)	204 (63.8)	211 (58)	136 (58)
No	1,213 (64)	71 (22)	112 (31)	70 (30)
Unknown	237 (12)	45 (14)	41 (11)	30 (11)
Median PSA at inclusion, ng/ml (IQR)	1.3 (0.70–2.5)	4.3 (3.5–5.6)	3.7 (2.7–5.2)	4.3 (3.2–6.0)
PSA category at inclusion, n (%)				
<1 ng/ml	719 (38)	0 (0)	0 (0)	0 (0)
1 to <2 ng/ml	568 (30)	0 (0)	30 (8.2)	12 (5.1)
2 to <3 ng/ml	246 (13)	0 (0)	87 (24)	37 (16)
3 to <4 ng/ml	123 (6.5)	123 (38)	81 (22)	51 (22)
4 to <5 ng/ml	87 (4.6)	87 (27)	66 (18)	48 (20)
5 to ≤ 10 ng/ml	110 (5.8)	110 (34)	100 (28)	88 (37)
>10 ng/ml	52 (2.7)	0 (0)	0 (0)	0 (0)
Unknown	0 (0)	0 (0)	0 (0)	0 (0)
5-ARI use at inclusion, n (%)				
Yes	42 (2.2)	9 (2.8)	13 (3.6)	8 (3.4)
No	1696 (89)	279 (87)	320 (88)	208 (88)
Unknown	167 (8.8)	32 (10)	31 (8.5)	20 (8.5)
Median STHLM3 score (IQR)	9 (5–15)	16 (11–24)	17 (13–24)	21 (17–28)
STHLM3 test result, n (%)				
Positive	364 (32)	247 (77)	364 (100)	236 (100)
Negative	769 (68)	72 (23)	0 (0)	0 (0)
Failed analysis	1 (0.09)	1 (0.31)	0 (0)	0 (0)
Highest PI-RADS score, n (%)				
1	26 (5.9)	21 (6.6)	19 (5.2)	11 (4.7)
2	294 (67)	211 (66)	237 (65)	142 (60)
3	8 (1.8)	4 (1.3)	7 (1.9)	2 (0.85)
4	72 (17)	51 (16)	65 (18)	48 (20)
5	37 (8.5)	33 (10)	36 (9.9)	33 (14)
1–2	320 (73)	232 (73)	256 (70)	153 (65)
3–5	117 (27)	88 (28)	108 (30)	83 (35)
ISUP grade group, n (%)				
Benign	27 (23)	21 (24)	21 (20)	14 (17)
1	52 (44)	37 (42)	50 (46)	37 (45)
2	29 (25)	22 (25)	28 (26)	23 (28)
3	4 (3.4)	4 (4.5)	4 (3.7)	4 (4.8)
4	5 (4.3)	4 (4.5)	5 (4.6)	5 (6.0)
5	0 (0)	0 (0)	0 (0)	0 (0)
≥ 2	38 (33)	30 (34)	37 (34)	32 (39)
Sepsis after biopsy, n (%)				
No	108 (92)	79 (90)	99 (92)	75 (90)
Yes	9 (7.7)	9 (10)	9 (8.3)	8 (9.6)

PSA = prostate-specific antigen; IQR = interquartile range; FDR = first-degree relative; PC = prostate cancer; 5-ARI = 5 α -reductase inhibitor; PI-RADS = Prostate Imaging-Reporting and Data System; ISUP = International Society of Urological Pathology.

MRI identified 117 participants (26.8%) with at least one PI-RADS 3–5 lesion, who then underwent MRGB. In total, 90 (77.0%) of the biopsied participants had PC, including 38 (42.2%) with GG ≥ 2 PC (Fig. 1).

The distributions of PI-RADS scores (1–5) and ISUP GG 1–5 did not significantly differ between STHLM311 positivity and PSA ≥ 3 ng/ml as the trigger for undergoing MRI ($p = 0.22$) and MRGB ($p = 0.24$; χ^2 test; Table 1).

There were nine (7.7%) MRGB-related cases of sepsis (Table 1).

3.2. Comparison of the STHLM3 and PSA tests

In comparison to PSA ≥ 3 ng/ml, a positive STHLM311 test increased detection of GG ≥ 2 PC from 30 to 37 cases (23.3%, 95% CI 5.6–52.2%; $p = 0.026$; Fig. 2A). A positive STHLM311 test also increased detection of GG 1 PC from 37 to 50 cases (35.1%, 95% CI 11.6–66.7%; $p = 0.006$) and led to

the same number of benign biopsies as PSA ≥ 3 ng/ml (21 vs 21; 0%, 95% CI –29.6% to 42.1%; $p = 0.53$; Fig. 2A). ROC curve analysis revealed that the STHLM3 test had an AUC of 0.78 for detection of GG ≥ 2 PC among the 437 men who underwent MRI, which was higher than the AUC of 0.63 for the PSA test ($p < 0.001$; Supplementary Fig. 1).

Furthermore, in comparison to PSA ≥ 3 ng/ml, a positive STHLM311 test increased the number of MRI scans from 320 to 364 (13.8%, 95% CI 0.9–21.7%; $p = 0.032$) and the number of MRGBs from 88 to 108 (22.7%, 95% CI 8.8–40.2%; $p = 0.003$; Fig. 2A). Thus, detection of seven additional GG ≥ 2 PCs (23.3% increase) according to STHLM311 positivity would also result in detection of 13 additional GG 1 PCs (35.1% increase) and require 44 additional MRI scans (13.8% increase) and 20 additional MRGBs (22.7% increase) in comparison to PSA ≥ 3 ng/ml (Supplementary Table 1). The STHLM3₁₁ test missed one GG 2 PC in the PSA interval 3–4 ng/ml and detected eight GG ≥ 2 PCs (seven GG 2 and one

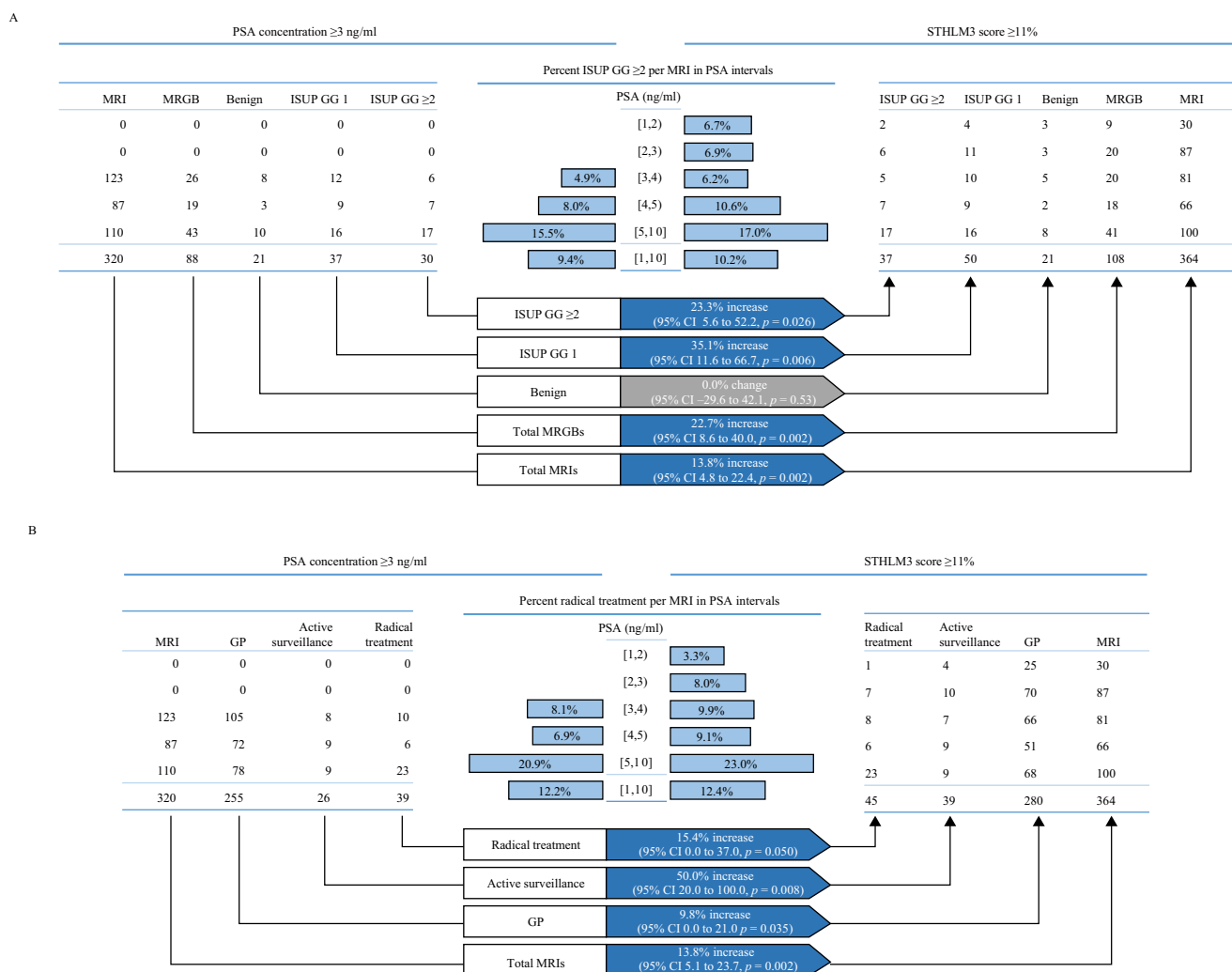


Fig. 2 – Comparison of PSA ≥ 3 ng/ml and the STHLM3 model with at least 11% risk for (A) detection of ISUP GG ≥ 2 prostate cancer and (B) clinical outcome. All patients who did not undergo radical treatment or active surveillance were discharged to the care of their GP. PSA = prostate-specific antigen; MRI = magnetic resonance imaging; MRGB = MRI-guided biopsy; ISUP GG = International Society of Urological Pathology 2014 grade group; GP = general practice; CI = confidence interval.

GG 4; [Supplementary Table 2](#)) in the PSA interval < 3 ng/ml at a cost of 107 additional MRI scans ([Fig. 2A](#)). In comparison to PSA ≥ 3 ng/ml, the STHLM3₁₁ test would increase the number of participants receiving radical treatment from 39 to 45 (15.4%, 95% CI 0.0–37.0%; $p = 0.050$) and the number receiving AS from 26 to 39 (50.0%, 95% CI 20.0–100.0%; $p = 0.008$; [Fig. 2B](#)).

3.3. Post hoc analysis for a higher STHLM3 cutoff

In the STHLM3-MRI screening study [18] the cutoff for STHLM3 positivity was adjusted post hoc from $\geq 11\%$ to $\geq 15\%$, as the latter detected the same number of GG ≥ 2 PCs as PSA ≥ 3 ng/ml in that population. In the current study, increasing the cutoff for STHLM3 positivity to $\geq 15\%$ (STHLM3₁₅) did not significantly change the number of GG ≥ 2 PCs (30 vs 32; 6.7%, 95% CI -8.1% to 25.9%; $p = 0.25$) or GG 1 PCs (37 vs 37; 0%, 95% CI -19.6% to 25.0%; $p = 0.52$) detected in comparison to PSA ≥ 3 ng/ml ([Fig. 3A](#)).

Use of STHLM3₁₅ would reduce the number of MRI scans from 320 to 236 (-26.2%, 95% CI -33.1% to -18.9%; $p < 0.001$)

and the number of benign biopsies from 21 to 14 (-26.3%, 95% CI -52.6% to 0.0%; $p = 0.034$; [Fig. 3A](#)). STHLM3₁₅ would miss two GG2 PCs in the PSA interval 3–10 ng/ml and detect four GG ≥ 2 PCs (three GG 2 and one GG 4 [Supplementary Table 2](#)) in the PSA interval < 3 ng/ml ([Fig. 3A](#)). There were no significant differences in the number of participants undergoing MRGB (88 vs 83; -5.7%, 95% CI -17.9% to 7.4%; $p = 0.20$; [Figure 3A](#)), radical treatment (39 vs 39; 0%, 95% CI -14.0% to 17.9%; $p = 0.53$), or AS (26 vs 27; 3.8%, 95% CI -20.6% to 36.8%; $p = 0.42$) between STHLM3₁₅ positivity and PSA ≥ 3 ng/ml ([Fig. 3B](#)).

In the current study, the STHLM3 cutoff that resulted in identical sensitivity to PSA ≥ 3 ng/ml for detection of clinically significant PC ($n = 30$) was a risk score $\geq 16\%$ ([Supplementary Table 1](#)).

4. Discussion

PRIMA is the first prospective clinical trial to directly compare the diagnostic performance of STHLM3 and PSA when

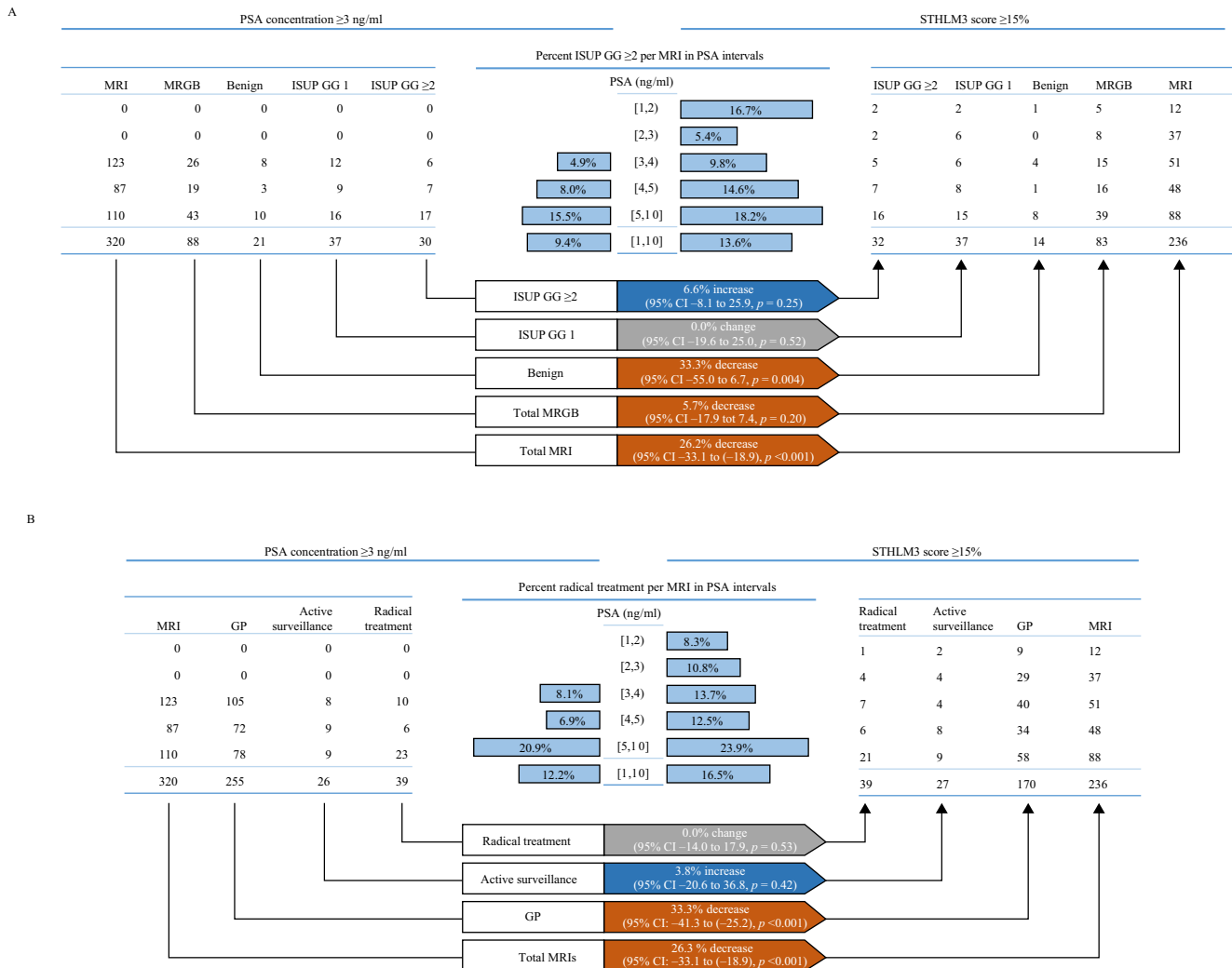


Fig. 3 – Comparison of PSA ≥ 3 ng/ml and the STHLM3 model with at least 15% risk for (A) detection of ISUP GG ≥ 2 prostate cancer and (B) clinical outcome. All patients who did not undergo radical treatment or active surveillance were discharged to the care of their GP. PSA = prostate specific antigen, MRI = magnetic resonance imaging, MRGB = MRI-guided biopsy, ISUP GG = International Society of Urological Pathology 2014 grade group, GP = general practice; CI = confidence interval.

offered in primary care as the entry test for prostate MRI. Thus, our study set-up corresponds to current clinical practice for PC detection in countries where systematic screening is not recommended [12–14,19]. While the STHLM3 test showed better diagnostic accuracy than PSA for detection of clinically significant PC, important questions remain regarding the clinical value of STHLM3 as an alternative or supplement to PSA for early PC diagnosis.

Our study complements two large screening-by-invitation studies from Sweden in which the STHLM3 test was developed and reported as noninferior to PSA ≥ 3 ng/ml for detection of GG ≥ 2 PC in comparison to systematic biopsy as the confirmatory test (STHLM3 study) [15] or biparametric MRI followed by systematic and targeted biopsy in MRI-positive cases (STHLM3-MRI study) [18].

In the PRIMA study cohort, use of STHLM311 positivity (score $\geq 11\%$) as the entry test for MRI increased detection of GG ≥ 2 PC by 23%, but also increased detection of GG 1 PC by 35% at the cost of 14% additional MRI and 23% additional MRGB procedures in comparison to PSA ≥ 3 ng/ml. This

is consistent with the recent STHLM3-MRI screening-by-invitation study, in which STHLM3 $\geq 11\%$ increased detection of GG ≥ 2 PC by 18%, GG 1 PC by 22%, MRI scans by 10%, and biopsies by 18% in comparison to PSA ≥ 3 ng/ml [18]. Together, these findings indicate that use of a STHLM3 cutoff of $\geq 11\%$ in the current diagnostic pathway or in a screening setting can increase detection of clinically significant PC. The increase in detection of clinically insignificant PCs with a positive STHLM3₁₁ test represents a major concern, as it is likely to lead to overdiagnosis, overtreatment, and additional downstream demands on MRI and urology services because of repeat examinations and medical consultations under the auspices of AS.

After initiation of the PRIMA study, data from the STHLM3-MRI trial became available, indicating that a higher cutoff risk (the STHLM3₁₅ test) could be beneficial [18]. Therefore, we performed a post hoc analysis in which we increased the cutoff for the risk score from 11% to 15%. The STHLM3₁₅ test retained sensitivity for GG ≥ 2 PC in the PRIMA cohort and detected the same number of GG 1 PCs

as use of PSA ≥ 3 ng/ml, and would avoid 26% of MRI scans and reduce benign biopsies by 33%. These results confirm and expand on findings from the STHLM3-MRI study, in which calibration revealed that STHLM3 $\geq 15\%$ had the same relative sensitivity for clinically significant PC as PSA ≥ 3 ng/ml, and would reduce MRI use by 36% and benign biopsies by 19% [18]. In comparison to use of PSA ≥ 3 ng/ml for MRI referral in the PRIMA study cohort, the higher risk cutoff would avoid 84 MRI scans, five MRGBs, and seven benign biopsies while maintaining sensitivity for GG ≥ 2 PC and not increasing the diagnosis of GG 1 PC. Notably, this would be at the added expense of 1134 STHLM3 tests (€250 per test) corresponding to the cost of 810 mpMRI scans (€350 per test, official tariffs in the Danish National Health Service [28]) in the current study. Thus, the cost for STHLM3 testing would have allowed us to perform mpMRI scans on 810 of the 814 (99.5%) men with PSA 1.0–2.9 ng/ml instead of a STHLM3 test. The optimal cutoff for the STHLM3 test will be a trade-off between sensitivity and specificity, as is also the case for the PSA test. The optimal cutoff may differ according to the specific setting in which the test is used. For example, in a population-based screening setting, very high specificity is more important than high sensitivity. Conversely, in a clinical setting in which men present with symptoms or relevant concerns, as in our study, high sensitivity may be more important than high specificity. Discussion of the pros and cons of different cutoffs for the STHLM3 test represents a space for patient involvement and shared decision-making. Although the STHLM3 test is associated with considerable added costs for blood analyses in comparison to the PSA test (€2 per PSA test), use of the STHLM3₁₅ test as an alternative to PSA—or an STHLM3 test with an even higher cutoff for positivity, as indicated in the [Supplementary material](#)—may still be beneficial if MRI capacity is limited. However, these alternative thresholds will require additional validation before potential implementation in clinical practice.

A strength of the present study is the prospective recruitment from general practice of a consecutive cohort of men from a large geographical area at a time when the MRI-first strategy had not been implemented in Denmark. Furthermore, all participants underwent both STHLM3 and PSA tests, thus avoiding the need for randomization. The participants had no other incitement for inclusion than what would normally prompt a PSA test from their GP, resembling current clinical practice in most countries, including the USA, UK, and Europe [12–14,19]. We recognize the smaller sample size in our study in comparison to the previous STHLM3-MRI study [18]. However, our sample size was sufficient to demonstrate significant differences in detection of GG 1 PC, GG ≥ 2 PC, and use of MRI and MRGB between the STHLM3 test and PSA.

A potential limitation is the use of a single observer for the MRI examinations. However, this observer was the only radiologist in the Central Denmark Region certified in prostate MRI by an international prostate MRI reference center (Radboud UMC, Nijmegen, The Netherlands). Similarly, an in-bore biopsy procedure was chosen as it was the only opportunity for MRI-guided targeted biopsy in the region during the entire study period. Sixty participants with either a PI-RADS score of 1–2 or negative MRGB subse-

quently underwent 10–12-core transrectal ultrasound-guided biopsy at a urology department. This revealed a single case of GG 2 PC, corresponding to a negative predictive value for GG ≥ 2 PC for mpMRI of 98.3%, in line with previous reports [7,10,11,29]. Together with the very low number of PI-RADS 3 lesions in the study (1.8% of 437 MRIs), this indicates high-quality MRI evaluation.

Another potential limitation of our study is that we used a previously established definition of clinically significant PC, ISUP ≥ 2 . However, this is in accordance with several previous diagnostic studies for PC [11,15,18,28,30] and thus also allows direct comparisons. Furthermore, we observed a discrepancy between the histopathological risk classification and treatment selection, whereby 14/49 patients (28%) with GG 1 PC received radical treatment and six of 29 patients (21%) with GG 2 PC were allocated to AS. This may reflect the contribution of MRI to clinical treatment decisions for GG 1 and GG 2 cancers, in addition to histopathological classification.

5. Conclusions

The increase in detection of clinically insignificant PCs using STHLM3₁₁ test positivity at the cost of greater use of MRI resources in comparison to PSA ≥ 3 ng/ml represents a major concern. While implementation of the STHLM3₁₁ test in a nonscreening workflow with MRI increased the sensitivity for detection of clinically significant PC, it did not increase the specificity.

Author contributions: Karina Dalsgaard Sørensen and Bodil Ginnerup Pedersen had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Study concept and design: Sørensen, Pedersen, Vedsted, Fredsøe, Borre.

Acquisition of data: Fredsøe, Sandahl, Jensen, Ulhøi, Borre, Sørensen, Pedersen.

Analysis and interpretation of data: Fredsøe, Sørensen, Pedersen.

Drafting of the manuscript: Fredsøe, Sandahl, Sørensen, Pedersen.

Critical revision of the manuscript for important intellectual content: Fredsøe, Sandahl, Sørensen, Pedersen.

Statistical analysis: Fredsøe.

Obtaining funding: Sørensen, Pedersen.

Administrative, technical, or material support: Sørensen, Pedersen.

Supervision: Sørensen, Pedersen.

Other (MRI evaluations): Pedersen.

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cancer patient'(#US10358681B2, #EP3262186B1, #ES2724404T3, #JP6769979B2), both licensed to Qiagen. The remaining authors have nothing to disclose.

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Appendix A. Members of the PRIMA Study Group

The following departments and investigators participated in the trial: Department of Urology, Regional Hospital Holstebro: Jørgen Bjerggaard Jensen, Lars Anders Høst, and Niels Mikkelsen; Department of Urology, Aarhus University Hospital: Michael Borre, Morten Heebøll Andersen, and Stefan Vind-Kezunovic; Department of Radiology, Aarhus University Hospital: Bodil Ginnerup Pedersen, Edvard Marinovskij, Mads Sandahl, and Tomas Frahm Nielsen; Department of Pathology, Aarhus University Hospital: Benedicte Parm Ulhøi; Research Unit for General Practice, Aarhus University: Ann Britt Krogh and Peter Vedsted; Department of Molecular Medicine, Aarhus University Hospital: Jacob Fredsøe and Karina Dalsgaard Sørensen.

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.euo.2023.07.006>.

References

- [1] Wilt TJ, Ahmed HU. Prostate cancer screening and the management of clinically localized disease. *BMJ* 2013;346:f325.
- [2] Hugosson J, Roobol MJ, Mansson M, et al. A 16-yr follow-up of the European Randomized study of Screening for Prostate Cancer. *Eur Urol* 2019;76:43–51.
- [3] Sung H, Ferlay J, Siegel RL, et al. Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 2021;71:209–49.
- [4] Moe A, Hayne D. Transrectal ultrasound biopsy of the prostate: does it still have a role in prostate cancer diagnosis? *Transl Androl Urol* 2020;9:3018–24.
- [5] Barry MJ, Simmons LH. Prevention of prostate cancer morbidity and mortality: primary prevention and early detection. *Med Clin North Am* 2017;101:787–806.
- [6] Johnson DC, Raman SS, Mirak SA, et al. Detection of individual prostate cancer foci via multiparametric magnetic resonance imaging. *Eur Urol* 2019;75:712–20.
- [7] Kasivisvanathan V, Rannikko AS, Borghi M, et al. MRI-targeted or standard biopsy for prostate-cancer diagnosis. *N Engl J Med* 2018;378:1767–77.
- [8] Siddiqui MM, Rais-Bahrami S, Turkbey B, et al. Comparison of MR/ultrasound fusion-guided biopsy with ultrasound-guided biopsy for the diagnosis of prostate cancer. *JAMA* 2015;313:390–7.
- [9] Wegelin O, Exterkate L, van der Leest M, et al. The FUTURE trial: a multicenter randomised controlled trial on target biopsy techniques based on magnetic resonance imaging in the diagnosis of prostate cancer in patients with prior negative biopsies. *Eur Urol* 2019;75:582–90.
- [10] Ahmed HU, El-Shater Bosaily A, Brown LC, et al. Diagnostic accuracy of multi-parametric MRI and TRUS biopsy in prostate cancer (PROMIS): a paired validating confirmatory study. *Lancet* 2017;389:815–22.
- [11] Hugosson J, Mansson M, Wallstrom J, et al. Prostate cancer screening with PSA and MRI followed by targeted biopsy only. *N Engl J Med* 2022;387:2126–37.
- [12] Mottet N, van den Bergh RCN, Briers E, et al. EAU-EANM-ESTRO-ESUR-SIOG guidelines on prostate cancer—2020 update. Part 1: screening, diagnosis, and local treatment with curative intent. *Eur Urol* 2021;79:243–62.
- [13] NICE guidance – prostate cancer: diagnosis and management. *BJU Int* 2019;124:9–26.
- [14] Carroll PH, Mohler JL. NCCN guidelines updates: prostate cancer and prostate cancer early detection. *J Natl Compr Cancer Netw* 2018;16(5 Suppl):620–3.
- [15] Grönberg H, Adolfsson J, Aly M, et al. Prostate cancer screening in men aged 50–69 years (STHLM3): a prospective population-based diagnostic study. *Lancet Oncol* 2015;16:1667–76.
- [16] Matuszczak M, Schalken JA, Salagierski M. Prostate cancer liquid biopsy biomarkers' clinical utility in diagnosis and prognosis. *Cancers* 2021;13:3373.
- [17] Vickers AJ, Russo G, Lilja H, et al. How should molecular markers and magnetic resonance imaging be used in the early detection of prostate cancer? *Eur Urol Oncol* 2022;5:135–7.
- [18] Nordström T, Discacciati A, Bergman M, et al. Prostate cancer screening using a combination of risk-prediction, MRI, and targeted prostate biopsies (STHLM3-MRI): a prospective, population-based, randomised, open-label, non-inferiority trial. *Lancet Oncol* 2021;22:1240–9.
- [19] Carter HB, Albertsen PC, Barry MJ, et al. Early detection of prostate cancer: AUA guideline. *J Urol* 2013;190:419–26.
- [20] World Medical Association. World Medical Association Declaration of Helsinki: ethical principles for medical research involving human subjects. *JAMA* 2013;310:2191–4.
- [21] Cohen JF, Korevaar DA, Altman DG, et al. STARD 2015 guidelines for reporting diagnostic accuracy studies: explanation and elaboration. *BMJ Open* 2016;6:e012799.
- [22] Fredsøe J, Koetsenruyter J, Vedsted P, et al. The effect of assessing genetic risk of prostate cancer on the use of PSA tests in primary care: a cluster randomized controlled trial. *PLoS Med* 2020;17:e1003033.
- [23] Turkbey B, Rosenkrantz AB, Haider MA, et al. Prostate Imaging Reporting and Data System version 2.1: 2019 update of Prostate Imaging Reporting and Data System version 2. *Eur Urol* 2019;2019(76):340–53.
- [24] Weinreb JC, Barentsz JO, Choyke PL, et al. PI-RADS Prostate Imaging-Reporting and Data System: 2015, version 2. *Eur Urol* 2016;69:16–40.
- [25] Sandahl M, Sandahl KJ, Marinovskij E, et al. Prostate cancer detection rate of manually operated and robot-assisted in-bore magnetic resonance imaging targeted biopsy. *Eur Urol Open Sci* 2022;41:88–94.
- [26] Epstein JI, Egevad L, Amin MB, Delahunt B, Srigley JR, Humphrey PA. The 2014 International Society of Urological Pathology (ISUP) consensus conference on Gleason grading of prostatic carcinoma: definition of grading patterns and proposal for a new grading system. *Am J Surg Pathol* 2016;40:244–52.
- [27] Hajian-Tilaki K. Sample size estimation in diagnostic test studies of biomedical informatics. *J Biomed Inform* 2014;48:193–204.
- [28] Risor BW, Tayyari Dehbari N, Fredsøe J, Sørensen KD, Pedersen BG. Cost-effectiveness analysis of Stockholm 3 testing compared to PSA as the primary blood test in the prostate cancer diagnostic pathway: a decision tree approach. *Appl Health Econ Health Policy* 2022;20:867–80.
- [29] Drost FH, Osses DF, Nieboer D, et al. Prostate MRI, with or without MRI-targeted biopsy, and systematic biopsy for detecting prostate cancer. *Cochrane Database Syst Rev* 2019;2019:CD012663.
- [30] de la Calle CM, Fasulo V, Cowan JE, et al. Clinical utility of 4Kscore®, ExosomeDx and magnetic resonance imaging for the early detection of high grade prostate cancer. *J Urol* 2021;205:452–60.